

PINAWA, MB, Canada

WMO: 714480

Lat: 50.177N

Lon: 96.065W

Elev: 879

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-23.1	-35.5	0.8	-28.7	-29.5	1.1	-22.8	18.8	7.7	16.7	6.6	2.4	180	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	84.8	69.2	81.7	67.3	79.0	65.5	72.5	80.8	70.1	78.5	67.9	75.9	6.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.7	112.9	77.5	67.1	103.0	74.8	64.8	95.0	72.4	36.8	80.5	34.7	78.5	32.8	76.1	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.3	12.5	DB	-36.5	91.1	5.1	3.2	-40.2	93.3	-43.1	95.2	-46.0	97.0	-49.7	99.3	(4)
(5)			WB	-36.4	76.5	5.0	3.0	-40.0	78.6	-42.9	80.4	-45.7	82.0	-49.3	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.1	2.1	6.6	21.4	37.6	50.5	61.3	66.1	64.3	55.4	41.8	25.0	10.7		
	DBStd	24.76	15.48	14.42	14.62	10.94	8.64	6.84	5.56	5.94	8.07	8.67	11.48	13.63		
	HDD50	6368	1484	1214	889	388	102	4	0	0	37	282	749	1219		
	HDD65	10404	1949	1634	1353	821	454	148	53	86	304	719	1199	1684		
	CDD50	1650	0	0	1	17	117	344	500	443	198	29	1	0		
	CDD65	209	0	0	0	0	4	37	89	63	15	1	0	0		
	CDH74	2102	0	0	0	5	99	417	813	622	138	8	0	0		
	CDH80	457	0	0	0	0	17	91	183	145	21	0	0	0		
(14) Wind	WSAvg	5.5	5.3	5.3	5.8	6.1	6.1	5.2	4.8	4.8	5.3	6.0	6.0	5.4		
(15) Precipitation	PrecAvg	21.6	0.9	0.6	1.0	1.3	2.4	3.4	3.1	2.8	2.2	1.7	1.2	1.1		
	PrecMax	30.9	2.6	1.8	3.3	3.0	5.2	7.7	6.8	6.4	4.5	4.1	3.5	3.0		
	PrecMin	13.5	0.1	0.0	0.1	0.4	0.6	0.7	0.4	0.5	0.4	0.3	0.3	0.3		
	PrecStd	4.3	0.5	0.4	0.7	0.6	1.3	1.8	1.4	1.4	1.1	1.1	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	41.4	59.2	73.2	82.5	87.8	89.1	88.2	82.9	73.7	57.2	40.0		
		MCWB	34.4	36.7	49.4	52.4	61.4	72.6	72.2	71.1	66.5	58.5	47.2	35.2		
	2%	DB	33.6	36.7	50.0	66.0	76.4	82.1	84.6	84.1	78.0	65.9	49.0	34.8		
		MCWB	31.4	33.9	43.1	49.5	57.2	66.5	70.2	68.5	64.4	54.9	42.1	32.2		
	5%	DB	29.3	33.2	44.6	60.7	71.7	78.7	81.6	80.6	74.0	60.5	43.9	31.9		
		MCWB	27.5	31.1	38.2	47.0	55.1	64.1	68.1	67.4	62.1	51.7	38.8	30.2		
	10%	DB	24.7	28.5	40.1	55.7	67.0	75.3	78.7	77.3	69.4	55.5	39.3	29.2		
		MCWB	23.2	26.4	35.4	44.4	53.8	62.9	66.5	65.7	59.9	49.0	35.7	27.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	37.9	50.3	55.0	64.7	76.0	75.6	75.1	70.8	60.1	48.5	35.7		
		MCDB	37.5	40.9	55.2	70.0	74.9	84.9	84.7	83.9	79.0	69.5	56.2	39.3		
	2%	WB	31.8	34.4	42.8	51.0	61.1	70.0	72.8	71.5	66.6	56.7	43.0	32.8		
		MCDB	33.6	36.6	48.8	63.3	69.7	78.6	80.9	80.1	73.0	64.4	48.3	34.7		
	5%	WB	27.7	31.2	38.7	48.1	58.2	67.0	70.5	69.2	64.0	52.9	39.2	30.3		
		MCDB	29.5	33.2	44.1	59.3	66.9	74.8	78.6	77.6	71.4	59.2	43.8	31.9		
	10%	WB	23.2	26.4	35.7	45.1	55.7	64.2	68.2	67.1	61.3	49.3	36.0	27.9		
		MCDB	24.6	28.2	39.7	55.1	64.8	71.6	76.1	74.8	67.8	55.1	39.3	29.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	21.3	21.1	21.5	21.8	20.1	20.3	20.8	19.5	15.4	13.2	15.4		
		MCDBR	17.6	19.3	22.1	29.8	30.8	25.9	24.1	26.0	26.4	24.8	19.6	17.2		
		MCWBR	15.8	16.1	15.2	16.8	15.5	13.0	12.2	13.2	14.7	14.9	13.4	14.9		
		MCDBR	17.2	17.1	20.5	26.5	23.6	20.9	21.2	22.4	21.4	22.3	18.2	16.3		
	5% WB	MCWBR	15.5	14.5	14.3	15.9	13.9	12.6	12.4	13.1	13.5	14.5	12.9	14.5		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.285	0.294	0.341	0.355	0.377	0.384	0.371	0.342	0.301	0.275	0.249		
	taud		2.274	2.201	2.341	2.388	2.416	2.381	2.381	2.430	2.498	2.576	2.484	2.339		
	Ebn at Noon		250	269	291	287	286	279	275	274	272	267	247	239		
	Edh at Noon		22	30	32	34	35	36	36	33	28	21	18	17		
(44) All-Sky Solar Radiation	RadAvg		397	769	1168	1579	1714	1805	1899	1571	1130	658	357	285		
	RadStd		42	57	90	121	137	145	95	114	97	87	47	31		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-2.00	-2.41	N/A	N/A	N/A	N/A	
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.28	N/A	N/A	N/A	N/A	

Nomenclature: See separate page